Contents

Chapter 2

Materials And Method

This chapter describes the experimental apparatus used in this work. First, a brief overview of the implementation of the Magneto-Optical Trap (MOT) for rubidium-87 atoms (^{87}Rb) is presented (this setup was already available prior to the present work). The chapter then focuses on the work carried out during this thesis, namely the implementation of the Optical Dipole Trap setup employed to transfer and trap atoms from the MOT. A particular focus is put on the coupling between the ODT laser source and the Hollow-Core Photonic Crystal Fiber (HCPCF), that delivers the ODT light in the vacuum chamber.

2.1 The Experimental Apparatus

In this section the experimental apparatus used to produce, cool, and trap ^{87}Rb atoms is presented. The preparation of the atomic sample relies on a well-established sequence consisting of a Magneto-Optical Trap followed by an optical molasses phase. These initial stages are essential to produce a cold atomic cloud, which provides the atoms for loading the Optical Dipole Trap which is the core addition and focus of this thesis work.

The MOT provides the initial spatial confinement and cooling of the background rubidium vapor. It operates by exploiting the scattering force exerted by three pairs of red-detuned, counter-propagating laser beams, spatially tuned by a quadrupole magnetic field gradient [1]. This stage effectively compresses the cloud and cools it down often already reaching the sub-Doppler regime. To further decrease the temperature and increase the phase-space density prior to the ODT loading, the magnetic field gradient is switched off, leaving the atoms exposed only to the cooling light in an optical molasses configuration. In this regime, lower sub-Doppler temperatures are achieved via Sisyphus cooling: a mechanism arising from the spatially varying polarization gradients created by the interference of the MOT beams, which forces the atoms to constantly climb optical potential hills, effectively draining their kinetic energy [1, 2].

2.1.1 Setup Description

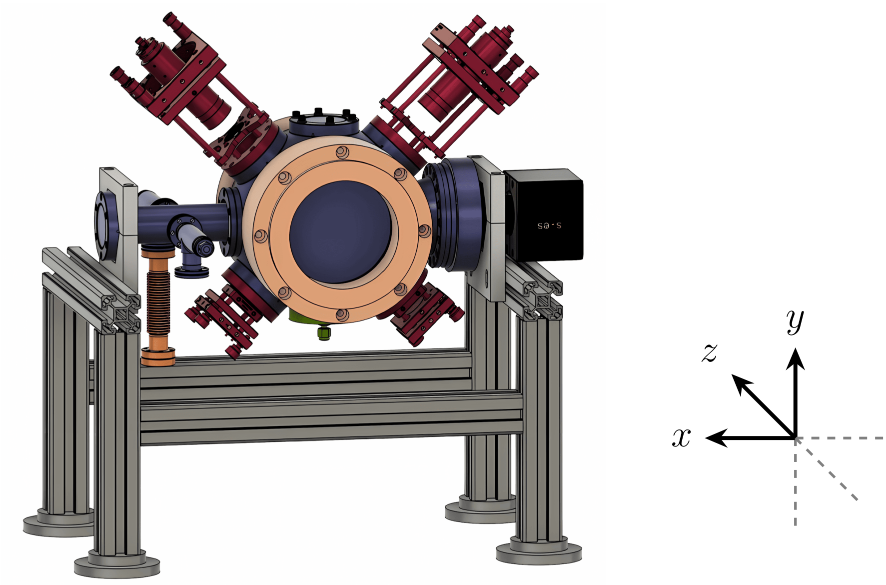

Figure 2.1: *CAD rendering of the vacuum chamber and surrounding experimental setup. The color coding highlights the main vacuum chamber (purple), MOT optics (red), MOT coils (orange), ion pump (black), rubidium source (dark orange, left), and the HCPCF port (green) located below the chamber. The compensation coils are not drawn. Credits [3, 4].*

The core component of the experimental apparatus is the vacuum chamber operating under Ultra-High Vacuum (UHV) conditions, inside which the atoms are trapped and cooled. The chamber is secured by a clamping structure to an optical table. A CAD rendering of the structure is displayed in Figure 2.1 together with the surrounding experimental elements.

The UHV is achieved through the employment of an internal pump (SAES Getter NEXTorr® Z200), composed of an active ion-pump and a passive non-evaporable getter (NEG). The internal pump is activated once the system reaches $\sim 10^{-7}$ mbar, which is initially attained using an external dual diaphragm-turbo pump (Pfeiffer Vacuum HiCube 80 Eco DN63). With the internal pump the system reaches a pressure of $\sim 10^{-10}$ mbar, as read from the ion-pump monitor.

The chamber is equipped with eight windows. Two large windows (mounted on CF100 flanges, diameter $\sim$ 100 mm) along the z direction, and the two pairs of smaller windows (mounted on CF40 flanges, diameter $\sim$ 40 mm) and oriented at 45° with respect to the optical table surface, are coated with 780 nm anti-reflective coating, to optimize the transmission of the six MOT beams. The remaining two CF40 windows are equipped with regular glass. One is mounted on the top side of the chamber (positive y-axis side), and the other on the positive x-axis side connected through a 4-way cross connector. The top window is utilized for the injection of the second beam of the ODT, whereas the x-axis window was intended for optical imaging in the chamber. However, this viewpoint directly faces the ion-pump facet (see Figure 2.1), introducing a significant parasitic background in the images due to reflection. For this reason, in this thesis work, the cameras were mounted on supports to observe the system from the two large windows on the z-axis ensuring a highly non-reflective background during image acquisition.

The rubidium vapor inside the chamber is provided by a 1 g sample of pure ^{87}Rb, connected through a valve on one of the four flanges of the 4-way cross connector, while the fourth flange is connected to the external vacuum pump. The rubidium sample is heated by a heating band powered by a variac. It is worth noting that opening the rubidium valve increases the pressure in the chamber from $\sim 10^{-10}$ mbar to $\sim 10^{-8}$ mbar.

The MOT also needs a magnetic field gradient to induce a spatially dependent Zeeman shift of the atomic energy levels. This shift induces a spatially dependent radiation pressure that pushes the atoms towards the magnetic field minimum. The magnetic field is provided by two circular coils (110 mm internal diameter) mounted on the $+z$ and $-z$ sides of the chamber. The coils are driven in an anti-Helmholtz configuration so that they create a quadrupole magnetic field. The quadrupole field is zero exactly midway between the coils (at the center of the chamber), and exhibits a constant gradient in its proximity. The measured axial gradient is 1.33 $\mathrm{G} \cdot \mathrm{cm}^{-1}/\mathrm{A}$. Due to Maxwell's divergence law $\nabla \cdot \mathbf{B} = 0$, the radial magnetic field gradient is half the axial gradient with opposite sign, conferring the typical ellipsoidal shape to the MOT cloud.

Furthermore, the experimental apparatus is provided by three pairs of compensation coils, commonly referred to as *shim coils*. These are square-shaped coils with an internal side of $\sim$ 650 mm, arranged in a mutually orthogonal configuration along the x, y, and z axes. Driven in a Helmholtz configuration, they produce a uniform magnetic field at the center of the chamber in any desired orientation. The compensation coils are primarily intended to cancel external magnetic fields, such as the Earth's one. However, they were also employed to shift the MOT position to increase its spatial overlap with the ODT beam.

The magnetic field generated by the compensation coils depends linearly on the applied current, with the following calibration factors [3]:

$$\begin{cases} B_z \approx 1.4\,\mathrm{mG/mA}, \\ B_x \approx 1.3\,\mathrm{mG/mA}, \\ B_y \approx 1.5\,\mathrm{mG/mA}. \end{cases} \tag{2.1}$$

On the bottom of the vacuum chamber (negative y-axis) a CF40 blind flange fitted with a 1/8" *Swagelok* feedthrough is mounted. A Teflon ferrule with a 0.5 mm diameter hole is inserted inside the Swagelok to host the HCPCF while ensuring proper sealing. Inside the chamber, the fiber is supported by a custom-designed mount to maintain strict linear

alignment, with its tip positioned approximately 10 mm below the chamber center.
Finally, the setup features three collimators (Schäfter + Kirchhoff, 60FC-Q780-4-M100S-37) that inject the MOT beams inside the chamber. These MOT beams pass through the windows and are retro-reflected by mirrors. In front of each mirror a $\lambda/4$ wave plate is placed so that the beams pass through it twice, ensuring that the two counter-propagating beams of each branch possess opposite circular polarization (σ^+ and σ^-). This polarization configuration is necessary for the realization of the MOT, and for the cooling. In this arrangement, the retro-reflected beams have an intensity of $\sim 88\%$ of the incoming ones. Although not perfectly balanced, the forces applied on the atomic cloud from opposite directions are similar enough to guarantee adequate trapping [3].
A key characteristic of this setup is that, except for the master laser locking system, the ODT injection line developed in this work, and the MOT beams, the entire apparatus is fully fiber-coupled. The system employs polarization-maintaining and absorption-reducing (PANDA) fibers [5] to ensure polarization stability. This approach offers several advantages: it minimizes misalignment caused by thermal drifts, makes the optical setups significantly more compact, and lowers safety risks associated with laser light propagating in free space. These features are particularly relevant in view of future integration of similar systems in other experimental platforms and for potential technology-transfer applications.

2.1.2 Laser System and Frequency Stabilization

The cooling system operates on the D_2 line transition $5^2S_{1/2} \rightarrow 5^2P_{3/2}$ of ^{87}Rb, which occurs at a wavelength of approximately 780 nm. To fully describe the cooling mechanism, it is necessary to consider the hyperfine structure characterized by the total angular momentum quantum number $F = I + J$. The hyperfine structure with the employed laser coupled transitions is shown in Figure 2.2 for both ^{85}Rb and ^{87}Rb.
The cooling transition is $F = 2 \rightarrow F' = 3$. At thermal equilibrium, prior to the illumination of the MOT beams, the ^{87}Rb atoms in the vapor are distributed between the $F = 1$ and $F = 2$ hyperfine ground states in a ratio of $3 : 5$, governed by their statistical weights. When the MOT beams are turned on the cooling laser starts to excite atoms to the $F' = 3$ state. According to the dipole transition selection rules $\Delta F = 0, \pm 1$, the only allowed decay is $F' = 3 \rightarrow F = 2$. Since in an absorption–emission cycle the atom starts and ends in the $F = 2$ state, the transition is referred to as a *closed transition*.
However, starting from $F = 2$, there is also a small probability that the atom undergoes the off-resonance excitation $F = 2 \rightarrow F' = 2$, and from $F' = 2$ the atom can spontaneously decay either to $F = 2$ or to $F = 1$. The latter is a *dark state* that does not interact with the cooling laser, therefore, an atom in $F = 1$ cannot be cooled or trapped anymore. To address this leakage, a repumping laser is introduced on resonance with the transition

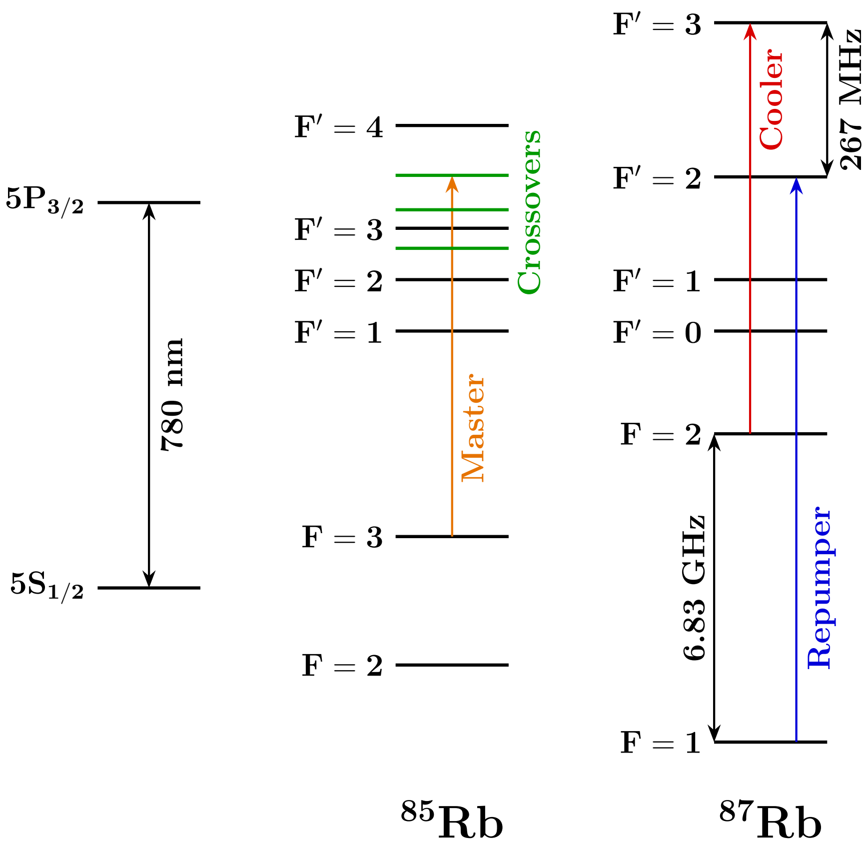

Figure 2.2: *Level scheme of the* ^{85}*Rb and* ^{87}*Rb* D_2 *line. Reference transitions are indicated for the master laser (orange), cooler laser (red), and repumper laser (blue). For* ^{85}*Rb, the crossover transition frequency used for saturation spectroscopy locking is also shown. Detailed data are available in [6, 7].*

$F = 1 \rightarrow F' = 2$, allowing the atoms to eventually decay back to $F = 2$ and re-enter the cooling cycle. The use of the repumper also brings the initial thermal population from the $F = 1$ state into the cooling cycle, maximizing the trap loading efficiency from the vapor.

In order to work properly, both the cooling and repumping lasers must be precisely frequency-locked to their target transition (with the appropriate detuning). The locking mechanism is briefly discussed in the following sections. A more exhaustive description of the laser system can be found in [8, 3].

2.1.2.1 Master laser

The cooling and the repumping lasers are locked to their target frequency by taking as a reference a third laser, known as the *master laser*. The master laser itself requires a precise frequency stabilization, which is achieved via *saturation spectroscopy* [9]. Saturation spectroscopy allows resolving spectral features much narrower than the Doppler width of a thermal vapor. For experimental convenience, the master frequency is chosen to lie between the cooling and repumping frequencies, Therefore, it is locked to the

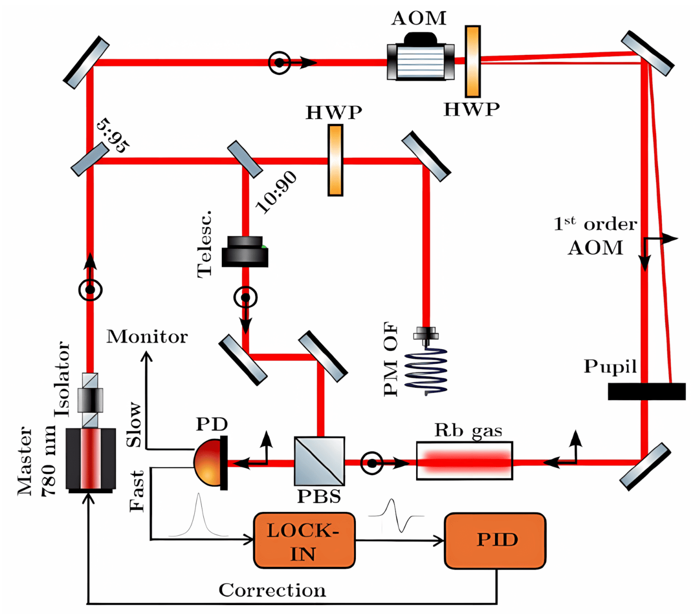

Figure 2.3: *Master laser locking setup scheme*

$(F = 3 \to F' = 3) \times (F = 3 \to F' = 4)$ crossover transition of the ^{85}Rb isotope (see Figure 2.2).

The locking scheme is illustrated in Figure 2.3. The light source is a 780 nm Distributed Feedback (DFB) laser diode (Toptica EYP-DFB-0780-00040-1500-BFW11-0005), current-driven and temperature-stabilized by custom electronics. The beam is split by a 5 : 95 beam splitter into two branches. The 5% branch takes the name of *probe beam*. The 95% branch encounters a subsequent 10 : 90 beam splitter: the 90% is routed to the cooler and repumper locking setup, while the 10% is used for the saturation spectroscopy and takes the name of *pump beam*.

The pump beam passes through a Polarization Beam splitter (PBS) where it is fully reflected before propagating through a ^{85}Rb heated vapor cell. On the other branch, the probe beam passes through an Acousto-Optic Modulator (AOM), driven by an amplified RF signal generated by a Direct Digital Synthesizer (DDS), controlled by the experiment electronics. Only the 1st diffraction order is retained, while the 0th order gets dumped by an iris. This AOM is used to frequency-modulate the probe beam as required by the locking protocol.

After the AOM, the probe beam passes through a half-wave plate (HWP) to adjust its polarization. It then traverses the rubidium cell, counter-propagating with respect to the pump beam. Upon exiting the cell, the polarized probe beam is fully transmitted by the aforementioned PBS and incident on a photodiode. The photodiode features two outputs: a slow port, connected to an oscilloscope for visual monitoring of the spectrum, and a fast port, whose signal is demodulated to extract the dispersive error signal. Finally, a PID controller processes this error signal to generate a feedback correction for the laser diode, actively stabilizing the laser frequency.

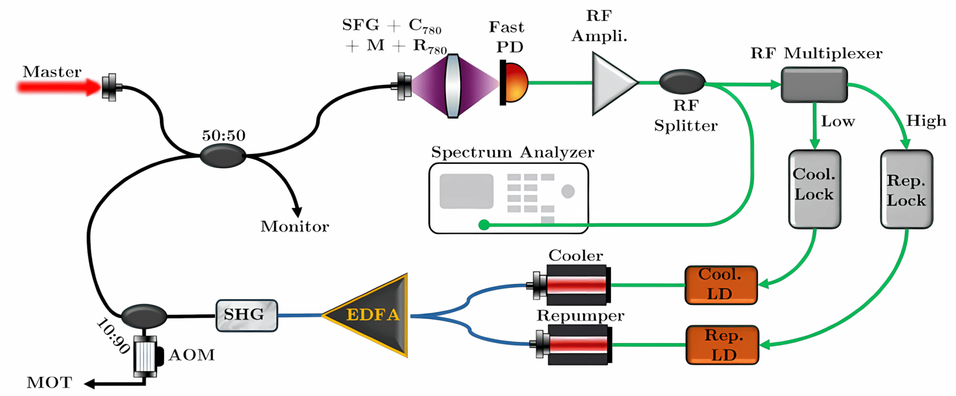

Figure 2.4: *Cooling laser and repumping laser setup scheme.*

2.1.2.2 Cooler and repumper lasers

The cooling and repumping light is generated by two 1560 nm DFB laser diodes (Eblana EP1562-0-NLW-B26-100FM) that are fiber-coupled and combined into a single fiber, which is then directed into an Erbium-Doped Fiber Amplifier (EDFA). Under standard operating conditions, the EDFA receives an input power of $\sim$ 5.2 mW, delivering an output of $\sim$ 500 mW, as read from its internal sensors.
Following the EDFA, the amplified light encounters a nonlinear crystal for Second Harmonic Generation (SHG), which converts the input wavelength from 1560 nm to 780 nm. The SHG crystal, together with the doubling of both cooling and repumping frequencies, also generates light at the sum frequency via *Sum-Frequency Generation* (SFG). This additional frequency is a byproduct and is not utilized in the experiment.
After the SHG stage, the light is split by a 10 : 90 beam splitter. The 90% fraction passes through an AOM, that acts as a fast switch, and goes to the three MOT beam collimators. The remaining 10% is combined with the Master laser light via a 2×2 fiber splitter to feed the cooler and repumper locking system. It should be noted that during this thesis work, the 10 : 90 splitter was damaged and was temporarily replaced by a 25 : 75 splitter.
The superimposed optical signal (composed of Master, Cooler, Repumper, and SFG frequencies) is detected by a fast photodiode with a high bandwidth of $\sim$ 10 GHz. The interference between these fields yields a signal that presents amplitude modulation at the beat note frequencies. In particular, the beat notes relevant for the frequency stabilization are the Master-Cooler beat at $\sim$ 1 GHz and the Master-Repumper beat at $\sim$ 5.5 GHz. The photodiode output signal is amplified and split into low- and high-frequency bands using RF filters to isolate the Cooler and Repumper beat signals. These signals are fed into two independent PID controllers, which drive the Laser Drivers (LDs) to actively stabilize the frequencies. This technique is known as *frequency offset locking*, or *beat note locking*.

2.2 Optical dipole trap

This section describes the implementation of the Optical Dipole Trap (ODT) setup used to transfer atoms from the MOT into a conservative optical potential. The ODT beam is guided through a Hollow-Core Photonic Crystal Fiber (HCPCF). As mentioned in Section 2.1, the HCPCF integrated into the experimental apparatus was already present prior to this thesis work. However, a test HCPCF was employed during the development of the setup in order to optimize the alignment procedure.
During the implementation of the system, dedicated diagnostic tools were also developed to investigate the near-field beam profile and improve the fiber-coupling alignment. The following sections describe the ODT optical configuration and the experimental procedures adopted for the alignment and loading of the trap.

2.2.1 1064nm laser characterization

The ODT is realized by using an IPG Photonics YLR-20-1064-LP-SF laser, a linearly polarized Ytterbium fiber laser that emits 1064 nm light, at a maximum nominal power of 20 W. For this laser model, the beam shape change significantly upon changes of the nominal power set. Therefore, the mode profile has been characterized as a function of the power set in order to allow a correct realization of the HCPCF injection setup. This calibration also provides a useful reference for future optimization of the HCPCF injection.

2.2.1.1 Methodology

To characterize the beam properties, its spatial profile is modeled as a Gaussian beam. Therefore, the beam intensity distribution can be described by [10]:

$$I(r,z) = I_0 \left(\frac{w_0}{w(z)} \right)^2 \exp\left(\frac{-2r^2}{w(z)^2} \right), \tag{2.2}$$

where

- z is the axial distance from the beam waist;
- r is the radial distance from the center of the beam;
- $w(z)$ is the beam radius, defined as the radius at which the intensity decreases of a factor $1/e^2$ with respect to the intensity at the beam center;
- $w_0 = w(0)$ is the beam waist, i.e. the beam radius at its focus;
- I_0 is the intensity at the center of the beam at its waist.

The transverse intensity profile follows a Gaussian shape, hence the name of Gaussian beam.

At any axial position z, the beam radius is given by [10]:

$$w(z) = w_0\sqrt{1 + \left(M^2\,\frac{z}{z_R}\right)^2} \tag{2.3}$$

where

$$z_R = \frac{\pi w_0^2}{\lambda} \tag{2.4}$$

is the Rayleigh range, which represent the characteristic distance around the waist over which the beam remains approximately collimated, and $M^2 \geq 1$ is the beam quality factor, accounting for deviations from an ideal Gaussian beam. By acquiring intensity images at multiple longitudinal positions and extracting the corresponding beam radii, Equation (2.3) can be fitted to determine both the waist radius and the waist position.

The experimental setup used for the intensity measurements corresponds to the initial section of the ODT injection setup (see Section 2.2.2.1). With reference to Figure 2.7, measurements were performed on the beam exiting the reflected branch of the second polarization beam splitter (PBS2). The power was adjusted through the regulation of the half-wave plates HWP2 and HWP3, together with the addition of optical density filters when needed. A Complementary Metal-Oxide-Semiconductor (CMOS) camera was employed to take 17 intensity images separated by 25 mm along the propagation axis for the nominal laser powers of 4 W, 7 W, 10 W, 13 W, 16 W and 19 W.

The straightforward approach of extracting the beam radius from each image consists of fitting a 2D asymmetric Gaussian profile and extract the radius along the x and y axis as

$$w_x(z) = 2\sigma_x, \qquad w_y(z) = 2\sigma_y. \tag{2.5}$$

However, fitting equation (2.3) using these extracted radii led to values of $M^2 < 1$ for all the power settings except 4 W, which is unphysical. However, enforcing the constraint $M^2 \geq 1$ resulted in visibly poor agreement between the fitted curves and the measured data. A likely explanation is that the presence of several optical components before the camera introduce aberrations or beam clipping that make equation (2.3) incorrect.

For this reason a second method called $D4\sigma$ was tested [11]. In this method, the beam radius is determined from the second moment of the beam intensity distribution with respect to the intensity centroid. For the x-axis:

$$\hat{\sigma}_x^2 = \frac{\int_{-\infty}^{\infty}(x - x_0)^2 I(x,y)\,dx\,dy}{\int_{-\infty}^{\infty} I(x,y)\,dx\,dy} \tag{2.6}$$

where x_0 is the intensity centroid x-coordinate, and $I(x,y)$ is the beam intensity profile.

Equation (2.3) is then fitted using the radii

$$w_x = 2\hat{\sigma}_x, \qquad w_y = 2\hat{\sigma}_y\,. \tag{2.7}$$

The $D4\sigma$ method should be more robust than the Gaussian fitting when dealing with non-ideal beam profiles, since it accounts for intensity contributions in the beam tails that may not be properly captured by a Gaussian fit. However, the $D4\sigma$ is very sensitive to anisotropic background noise, which can bias both the centroid and second-moment calculations. In the present setup, the presence of the PBS1 and the beam dump that dissipates a large quantity of optical power, generates a non-uniform scattering background. This asymmetry introduces systematic distortions in the $D4\sigma$ analysis. Consequently, even this method did not consistently yield $M^2 \geq 1$ without constraints, although it performed better than the Gaussian fit, giving naturally $M^2 \geq 1$ for 4 W, 16 W and 19 W.

For both analysis methods, a preliminary Gaussian fit was first performed to find approximately the beam center position. A region of interest (ROI) extending to 5σ around the beam center was then selected. Subsequently, either a refined Gaussian fit was performed or the centroid and second moments were calculated to determine the beam radius before fitting Equation (2.3).

2.2.1.2 Results

The beam waist radii and waist positions obtained with the two analysis methods are summarized in Tables 2.1 and 2.2. The fit of Equation (2.3) was performed using the Trust Region Reflective (TRF) algorithm while imposing the constraint $M^2 \geq 1$. The final reported values were obtained by averaging the results of the Gaussian fit and $D4\sigma$ methods.

The waist radius as a function of the nominal laser power is plotted in Figure 2.5 top. Except for 4 W and 7 W, the results reveal a small astigmatism, with the waist measured along the x direction exceeding that along the y direction by approximately 1–2%. Furthermore, the graph shows that the waist radius decreases approximately linearly from 7 W to 19 W, indicating a significant power dependence of the laser mode size.

The waist longitudinal position as a function of the nominal laser power is plotted in Figure 2.5 bottom. These values should be interpreted with caution. Imposing $M^2 \geq 1$ allow the fits to better match the middle and far points which depends asymptotically on solely w_0 (limit $z \gg 1$ in equation 2.3), but strongly miss the prediction on the point at short distance that give the information on the waist position (see Figure 2.6). Although the extracted values carry significant systematic uncertainty, the observed trend appears physically plausible, with the beam waist shifting downstream as the nominal laser power increases from 4 W to 13 W.

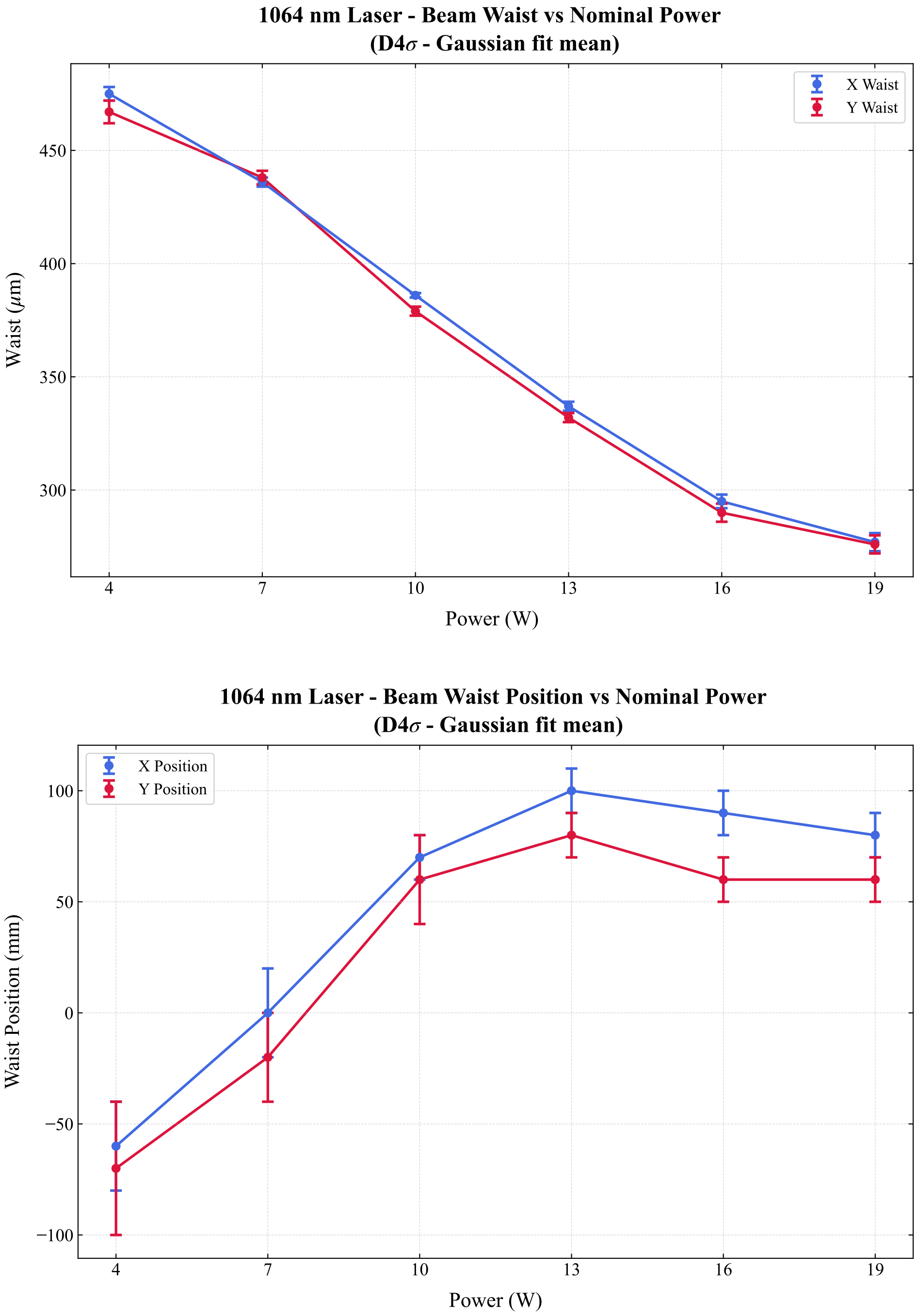

Figure 2.5: *Characterization of the 1064 nm laser beam parameters as a function of nominal output power. (Top) Transverse beam waist size versus power. (Bottom) Longitudinal position of the waist. The reported values correspond to the average of the results obtained using the D4σ and Gaussian-fit methods. Blue and red markers denote measurements along the orthogonal x and y transverse axes, respectively.*

Table 2.1: *Comparison of the beam waist w_0 obtained via the D4σ method and the Gaussian fit method. The final values correspond to the average of the two estimates.*

Nominal Power (W)	Estimated waist w_0 (μm)					
	D4σ-x	*D4σ-y*	Gaus-x	Gaus-y	Avg-x	Avg-y
4	481 ± 5	474 ± 7	469 ± 4	460 ± 7	475 ± 3	467 ± 5
7	454 ± 3	452 ± 4	419 ± 3	423 ± 5	436 ± 2	438 ± 3
10	403 ± 2	394 ± 3	370 ± 2	364 ± 3	386 ± 1	379 ± 2
13	351 ± 2	343 ± 3	323 ± 4	321 ± 4	337 ± 2	332 ± 2
16	307 ± 2	298 ± 4	283 ± 6	282 ± 6	295 ± 3	290 ± 4
19	288 ± 3	284 ± 4	266 ± 7	268 ± 6	277 ± 4	276 ± 4

Table 2.2: *Comparison of the beam waist position z_0 extracted via the D4σ method and the Gaussian fit method. The final values correspond to the average of the two estimates.*

Nominal Power (W)	Estimated waist position z_0 (mm)					
	D4σ-x	*D4σ-y*	Gaus-x	Gaus-y	Avg-x	Avg-y
4	-90 ± 40	-80 ± 50	-40 ± 20	-60 ± 40	-60 ± 20	-70 ± 30
7	20 ± 30	-3 ± 40	-20 ± 20	-40 ± 20	0 ± 20	-20 ± 20
10	90 ± 20	70 ± 30	50 ± 10	40 ± 10	70 ± 10	60 ± 20
13	110 ± 10	90 ± 20	90 ± 20	70 ± 20	100 ± 10	80 ± 10
16	72 ± 9	50 ± 20	100 ± 20	80 ± 20	90 ± 10	60 ± 10
19	70 ± 10	40 ± 10	90 ± 20	70 ± 20	80 ± 10	60 ± 10

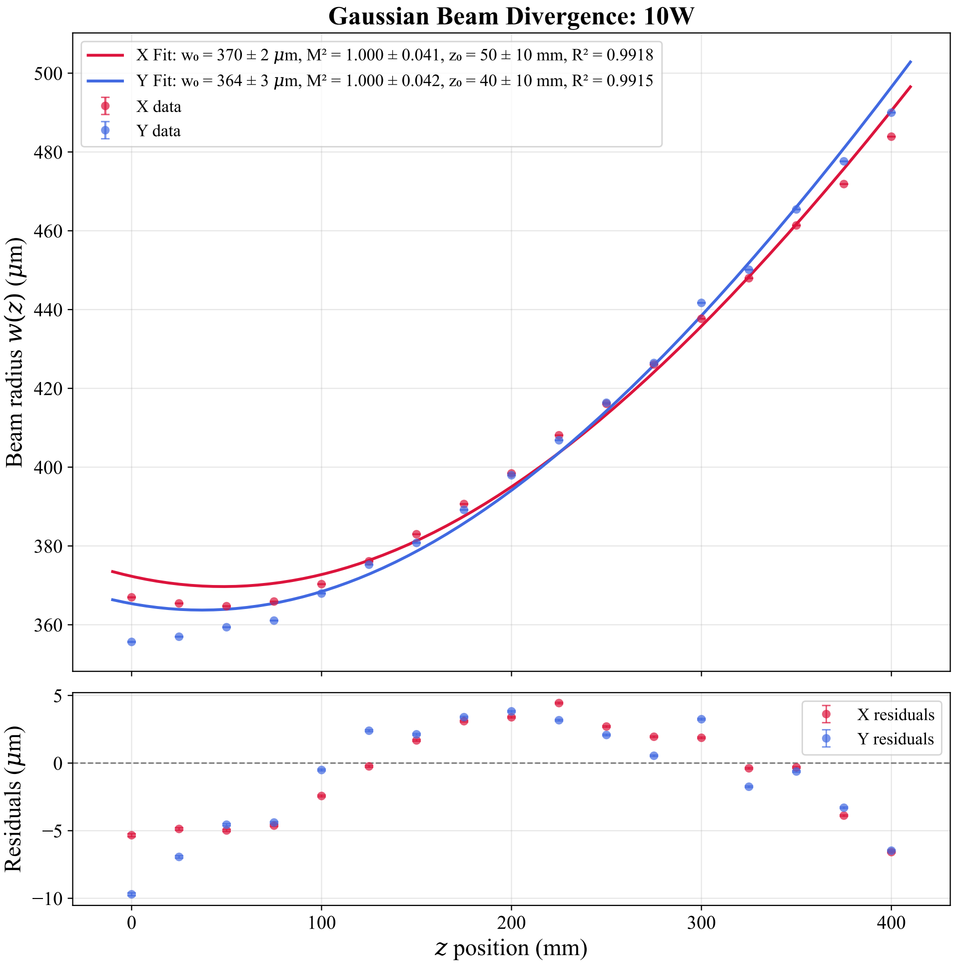

Figure 2.6: *Gaussian beam divergence profile measured at a nominal laser power of 10 W. The beam radius $w(z)$ is shown as a function of the longitudinal position z along the x (red) and y (blue) axes. Solid lines denote Gaussian propagation fits performed under the constraint $M^2 \geq 1$. The systematic discrepancy observed at short propagation distances ($z < 100mm$) results from the imposed M^2 constraint, which prioritizes agreement with the intermediate- and far-field data at the expense of the near-field region. Consequently, the accuracy of the fitted beam waist position z_0 is reduced.*

2.2.2 Optical setup

In this experiment the ODT light is guided through a HCPCF, which introduces a critical complication in the injection procedure. Indeed, in contrast with standard telecom fibers, HCPCFs are multi-mode fibers. As a consequence, even small deviations of the injected beam from the ideal injection angle excites higher order modes in addition to the fundamental one. This results in distorted output beam profiles, which are not well suited for the realization of an optical conveyor belt based on the superposition of the beam guided by the HCPCF and a beam guided through a conventional telecom fiber supporting only the fundamental LP_{01} mode.
For this reason, particular cure was devoted to the alignment procedure. A significant improvement in the quality of the transmitted mode was achieved by optimizing the injection while inspecting the near-field of the transmitted beam, rather than simply maximizing the transmission efficiency. The following sections describe the injection setup and the horizontal microscope built for the near-field inspection.

2.2.2.1 Injection setup

Figure 2.7 shows the optical setup scheme used for the injection of the laser light in the HCPCF. The scheme is composed of the following elements:

- **Half Wave Plate 1** (HWP1) and **Optical Isolator**. The optical isolator prevents back reflections from propagating towards the laser cavity. High-power lasers are particularly sensitive to back reflections, which can introduce power instabilities and potentially damage the laser system. Since optical isolators operate correctly only for a specific input polarization, HWP1 is used to optimize the polarization of the incoming beam and maximize the transmission through the isolator.

- **HWP2** and **Polarization Beam Splitter 1** (PBS1). These components are used to regulate the optical power delivered to the subsequent stages of the setup. The unwanted fraction of the beam is directed onto a beam dump. This configuration is particularly useful during alignment procedures, where the optical power is reduced to a few milliwatts for safety reasons.

- **HWP3** and **PBS2**. PBS2 splits the beam into two different branches. The first is employed for the HCPCF injection, while the second will be injected into a PANDA polarization-maintaining fiber, which eventually will provide the second beam for the realization of the optical conveyor belt. HWP3 is adjusted to have approximately 50/50 power splitting ratio, although this value can be modified if required.

- **Acousto-Optic Modulator** (AOM). The AOM is employed as a fast switch for the ODT beam. The device operates using an acoustic wave at 110 MHz, generated by a

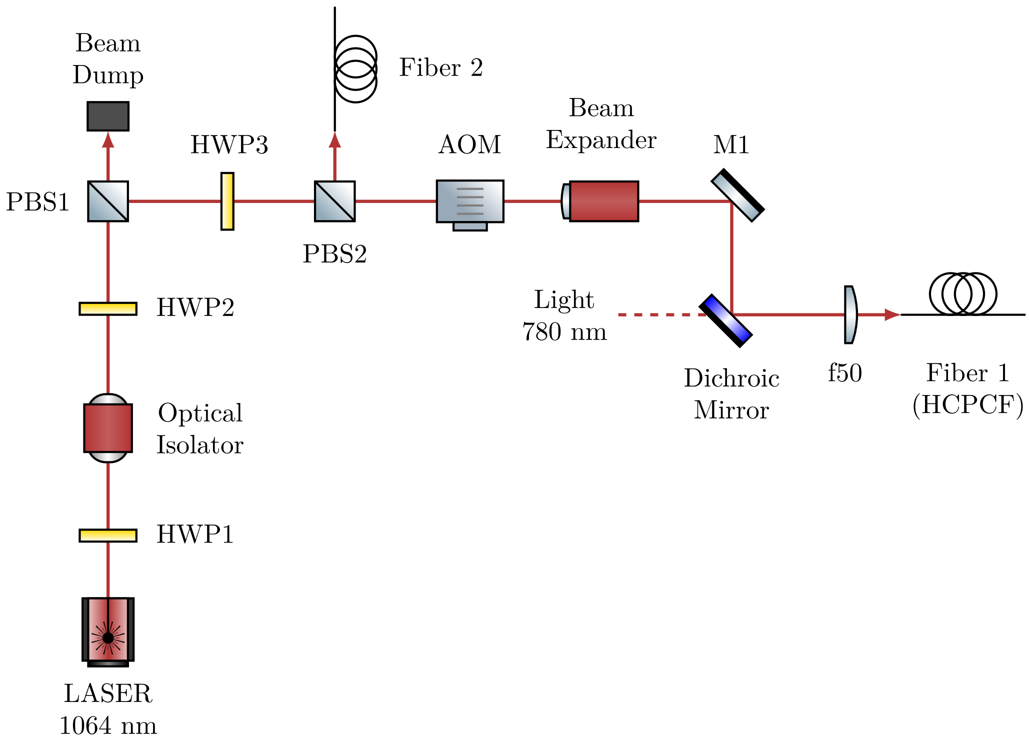

Figure 2.7: *Scheme of the optical setup for 1064 nm ODT beam injection into the HCPCF. The Acousto-Optic Modulator (AOM) operates at 110 MHz, and the subsequent setup utilizes the fist diffraction order. The zero-th order is blocked by a beam dump (not drawn). The 780 nm light setup is not drawn, its presence will be important in the future stages of the experiment for the characterization of the atoms inside the HCPCF core. Fiber 2 injection setup is not drawn as well.*

Direct Digital Synthesizer (DDS), causing the first diffraction order to be frequency shifted by the same amount. Although this shift is negligible with respect to the trap depth of the ODT, it is essential for the realization of the optical conveyor belt.

- **Beam expander**. A ∼3X beam expander is employed to increase the beam diameter and collimate it before the focusing conditions for coupling into the fiber.

- **Mirror 1** (M1) and **Dichroic Mirror**. Both mirrors are mounted on tilt stages and their angular adjustment is used for optimizing the fiber injection alignment. The dichroic mirror also combines the 1064 nm ODT beam with a 780 nm beam that will be used for the characterization of the atoms trapped inside the HCPCF core.

- **Focusing Lens** (f50). A lens with focal length $f = 50$ mm focuses the beam into the HCPCF. It is mounted on a horizontal translation stage, providing an additional degree of freedom for optimizing the injection alignment.

- **Fiber 1** (HCPCF). The HCPCF is mounted on an x-y translation stage and terminated with an FC/APC connector. The translation stage allows transverse displacement of the fiber, providing additional alignment degrees of freedom during the injection procedure, although these degrees of freedom were found to have limited impact on the final optimization.

Using the beam waist measurements in Section 2.2.1.2, a nominal laser power of 10 W was selected as the optimal operating point. Since the fitted waist positions were found to be affected by large systematic issues, the beam waist was assumed to be located at the output face of PBS2. The radius at the entrance of the beam expander can be estimated using (2.3), and the measured distance of approximately 175 mm between PBS2 and the beam expander (see figure 2.9). Assuming an ideal beam $M^2 = 1$, using the x waist value for 10 W of $w_0 = 384$ μm, the beam radius at the entrance of the beam expander is estimated to be w(beam expander entrance) $\simeq 414$ μm.
The beam expander provides a magnification factor of three, yielding w(beam expander exit) $\simeq$ 1240 μm. At this stage the corresponding Rayleigh range, calculated using 2.4 is $z_R \simeq$ 4.5 m. Since the remaining propagation distance before the injection optics is negligible compared to the Rayleigh range, the beam is considered approximately collimated. Under this assumption, the beam waist after the injection lens can be estimated using [10]:

$$w_{\text{out}} = \frac{\lambda f}{\pi w_{\text{in}}}, \tag{2.8}$$

where $f = 50$ mm is the focal length of the injection lens. Propagating the uncertainty, one gets:

$$w_{\text{inj}} \simeq (13.6 \pm 0.1)\ \mu\text{m}.$$

This result is obtained in the ideal case, the real waist at the injection is expected to be closer to the target value of approximately ~ 18 μm. For instance, assuming a more realistic beam quality factor of $M^2 = 1.2$ leads to

$$\tilde{w}_{\text{inj}} \simeq (15.1 \pm 0.1)\ \mu\text{m}$$

Additional deviations from the ideal prediction are expected due to the AOM, which distorts the beam profile by compressing it along the y direction and expanding it along the x direction, resulting in an expected beam radii ratio $w_y/w_x \approx 0.85$ at AOM exit, which is reversed at injection by the relation (2.8).
To verify these predictions, the beam profile after the injection lens was characterized following a procedure analogous to that described in Section 2.2.1, yielding

$$w_{\text{inj-}x} = (17.1 \pm 0.7\ \mu\text{m}) \qquad w_{\text{inj-}y} = (19.3 \pm 0.4\ \mu\text{m}),$$

sufficiently close to the target waist.

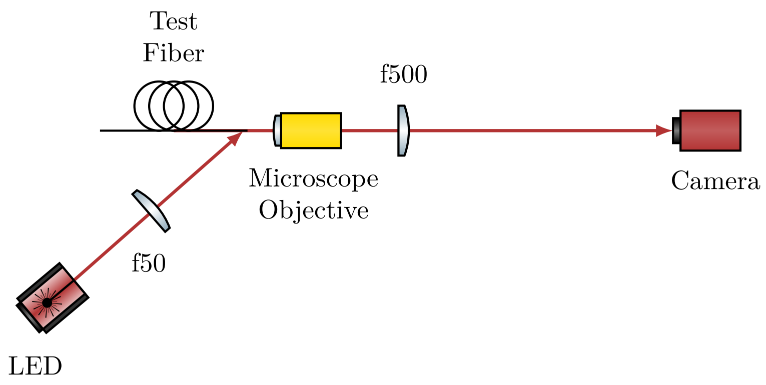

Figure 2.8: *Scheme of the optical setup for the horizontal microscope.*

2.2.2.2 Horizontal microscope

Initially, the injection into the HCPCF installed inside the vacuum chamber was attempted by using a Complementary Metal-Oxide-Semiconductor (CMOS) camera to monitor the beam profile. However, despite the effort, strongly deformed mode profiles were obtained.

In order to better investigate the injection procedure and simplify the alignment process, a separate test HCPCF was employed outside the vacuum chamber. The test fiber consisted of an approximately 50 cm HCPCF segment with the same FC/APC connector as the fiber in the chamber. The test fiber allowed an easy measurement of the efficiency on the exit facet of the beam without any window in the path, and being outside the vacuum chamber allow for easier mode imaging. Moreover, since the two fibers shared the same connector, the optimal alignment conditions identified with the test fiber were expected to provide a good starting point for the alignment of the HCPCF installed inside the chamber. Nevertheless, even with the test fiber, optimization based solely on transmission efficiency and far-field inspection still resulted in poor beam quality.

The fiber manufacturer suggested that the optimal injection condition should be determined by inspecting the shape of the near-field of the transmitted mode. For this purpose, a horizontal microscope system was developed to inspect the HCPCF facet and to investigate the near-field profile of the transmitted mode. The optical scheme of the microscope is shown in Figure 2.8.

The fiber was positioned inside a 90° V-groove of depth 350 μm. The fiber was fixed inside the groove by means of a thin foam layer attached to a tiltable mount using adhesive tape. The coating was removed from the fiber end, leaving approximately 15 mm of suspended fiber in order to allow lateral illumination of the facet.

The HCPCF facet was illuminated by a red LED (wavelength $\lambda = 635$ nm) focused by an $f = 50$ mm lens onto the final suspended portion of the fiber. Part of the red light propagated through the silica structure forming the kagome lattice, making them visible on the microscope camera. The observation of the kagome structure was used to optimize

the microscope focus and to inspect the integrity and cleanliness of the fiber facet.
The light emerging from the fiber was collected by an optical microscope objective lens (Olympus PLN10X objective, nominal focal length $f = 18$ mm, working distance $d = 10.6$ mm) and subsequently re-imaged onto a CMOS camera by means of an $f = 500$ mm lens. The expected magnification is $M = f_{\text{tube}}/f_{\text{lens}} \approx 27.8$. The objective is mounted on a horizontal translation stage used for the focussing. The camera was connected to a computer for real-time imaging of the transmitted near-field mode. In addition, a powermeter could be inserted between the fiber output and the objective lens in order to evaluate the coupling efficiency by comparing the transmitted power with the optical power measured before the injection stage. Finally, a black paper tube was placed between the imaging $f = 500$ mm lens and the camera to reduce background light from the LED illumination and from ambient room light.
The alignment procedure was carried out by optimizing the transmitted near-field profile in order to obtain a mode as close as possible to a single-lobed structure, results and further details are found in Section 2.2.3.
Figure 2.9 shows the top-down photographs of the complete experimental apparatus, including both the injection setup and the horizontal microscope system, and Figure 2.10 shows a detailed view of the horizontal microscope.

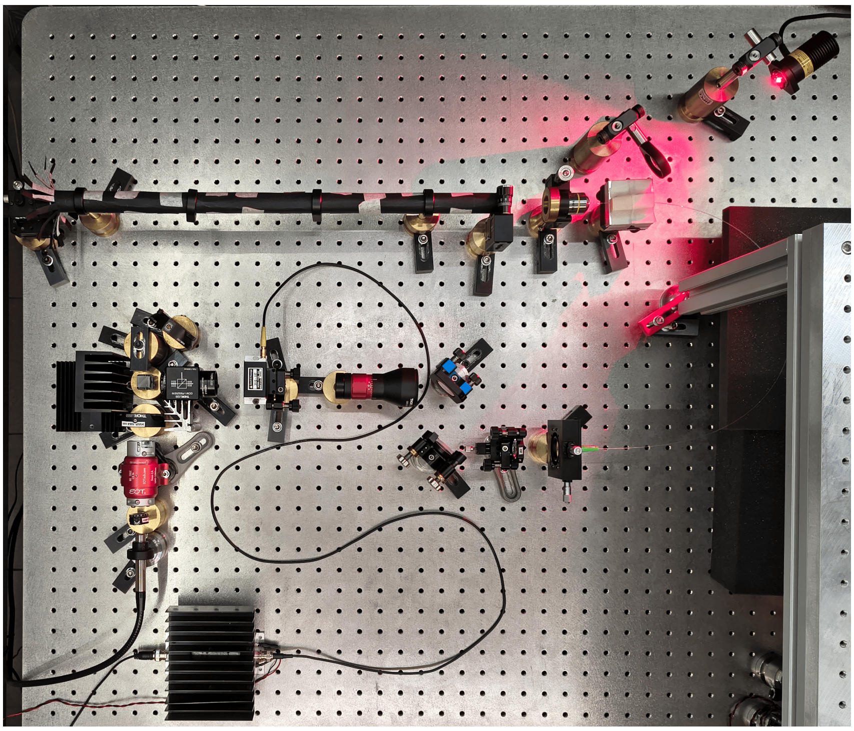

Figure 2.9: *Top-down photograph of the complete experimental apparatus. The horizontal microscope, shielded by the black paper tube, is visible at the top of the image. The optical injection setup is located in the center, while the RF amplifier driving the AOM is placed at the bottom. For reference, the distance between the screw holes on the breadboard is 25 mm.*

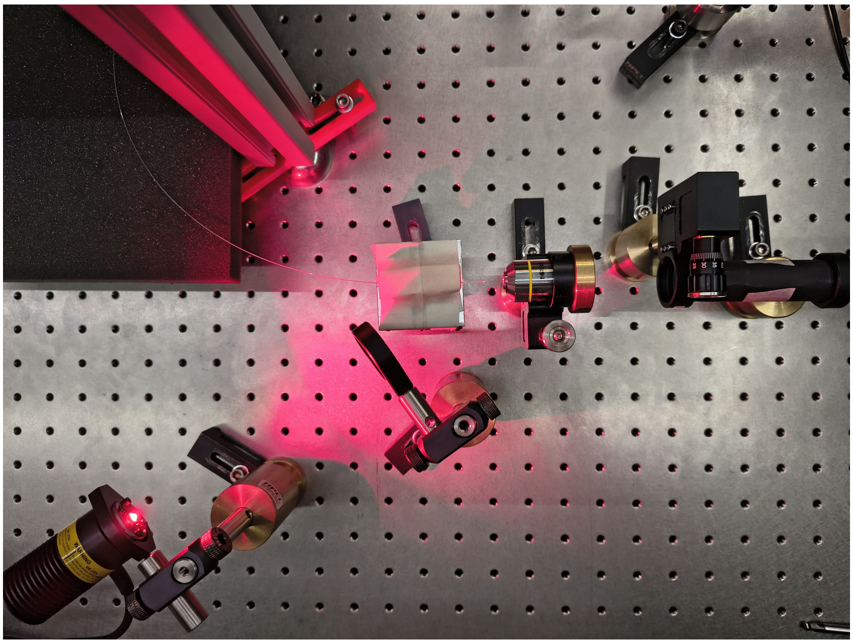

Figure 2.10: *Detailed top-down photograph of the horizontal microscope. The setup highlights the test fiber, the red LED providing lateral illumination to the fiber facet, the microscope objective lens and the refocusing lens.*

2.2.2.3 Chamber fiber optical imaging setup

After the optimization and the validation of the injection alignment procedure using the test fiber, the same approach was applied to the fiber inside the vacuum chamber.

To investigate the near-field profile, a dedicated optical imaging setup was build on a breadboard positioned above the vacuum chamber (Figure 2.11). Since the optical fiber is located inside the chamber, a first stage microscope was designed to create a virtual image of the fiber facet outside the chamber. For this purpose a f150 lens ($f = 150$ mm) was placed approximately 150 mm above the fiber facet, directly above the chamber window, such that the light emerging from the fiber is collimated. The collimated beam propagate through an opening and is redirected horizontally by a mirror (M1) tilted at 45°. A second lens f300 ($f = 300$ mm) refocuses the beam, forming an inverted image of the fiber facet with a magnification of $\times 2$.

A second imaging stage, consisting of lenses L3 and L4, acted as a microscope for this intermediate image and projected a magnified image onto the sensor of a camera. The focal lengths of L3 and L4 were modified several times throughout the experiment to optimize the field of view and magnification. The configuration used most frequently employed an L3 lens with $f = 100$ mm and an L4 lens with $f = 500$ mm, resulting in an additional magnification of $\times 5$. Combined with the first imaging stage, this yielded an overall magnification of approximately $\times 10$, which was used for the majority of the measurements.

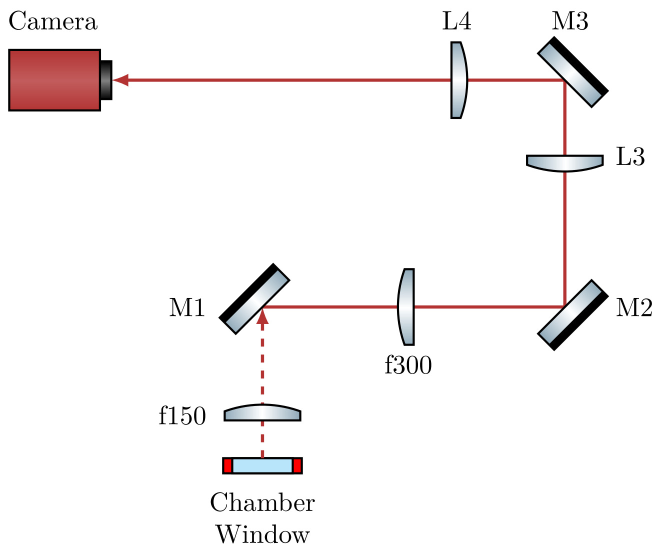

Figure 2.11: *Scheme of the optical setup used to image the fiber inside the vacuum chamber. The optical path between the chamber window and the mirror M1 is vertical (characterized by the dashed beam line), whereas the rest of the setup is arranged horizontally on a breadboard mounted above the chamber. The focal length of lenses L3 and L4 were varied several times during the experiment. Most images where acquired using an* $f = 100$ mm *and an* $f = 500$ mm *providing an overall magnification of approximately* $\times 10$ *combined with the first imaging stage.*

2.2.3 Injection alignment and near-field imaging

For a conventional single-mode telecom fiber, the alignment procedure is straightforward: it is sufficient to inject a low-power beam of few milliwatts, monitor the transmitted power at the fiber exit, and adjust the injected beam to maximize the guided optical power.

By contrast, aligning the injection of a multimodal HCPCF is significantly more complex. The silica jacket that surrounds the kagome core can guide the light with coupling efficiencies as high as 60%, although with badly shaped mode (see figure 2.12). Considering also that the jacket is large compared to the core, monitoring just the coupling efficiency is particularly ineffective, as the optimization can easily get trapped in a local maximum associated with jacket guiding.

As suggested by the manufacturer, the best procedure for the injection alignment resulted to be monitoring the near-field mode shape at the fiber exit. The complete alignment procedure begins with the use of a powermeter to ensure the impinging beam hits the fiber, even on the jacket. Then switch to live monitoring of the near-field with a camera and optimize its shape by tuning the degrees of freedom of the injection setup. Coupling efficiency is measured only after the mode profile has been optimized.

Applying this protocol, the test fiber injection was optimized employing the injection optical setup and the horizontal microscope described in Section 2.2.2. This approach yielded to a good result and thus was subsequently applied to align the injection into the

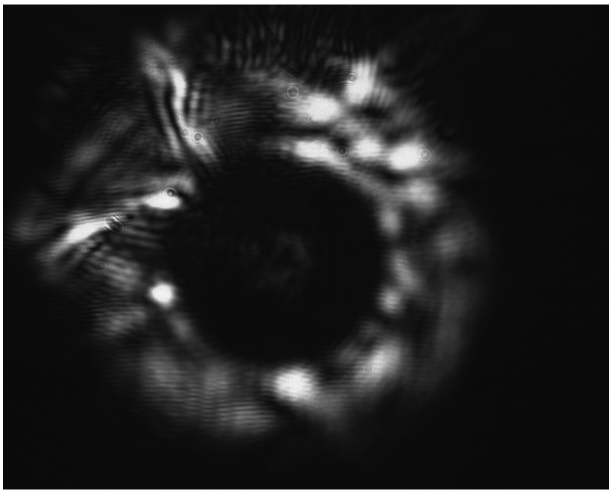

Figure 2.12: *Example of the near-field of a mode guided by the test fiber jacket. The light is mainly guided by the silica jacket surrounding the kagome structure, while only a weak signal is visible in the core. Note that this strongly deformed mode had a coupling efficiency of* $\sim 60\%$.

fiber in the vacuum chamber.

2.2.3.1 Test fiber

The test fiber was cleaved prior to inspection. Since the horizontal microscope has a sufficient resolution to resolve the kagome structure of the fiber, it was exploited for the focusing procedure. By monitoring the camera imaging in real-time, the kagome structure was focussed using the translation stage of the microscope objective. The objective refractive index can be considered roughly constant across the visible spectrum and into the near-infrared (NIR). This guarantees that when the structure (illuminated by red light) is in focus, the guided mode (1064 nm NIR light) is also in focus. Figure 2.13 shows an image of solely the kagome structure alongside an image of the kagome structure together with some guided light. The image show that the kagome structure is intact confirming that the cleaving was correctly performed. Furthermore, the mode is clearly guided by the core and exhibits a single lobe, however, the shape is not symmetric with respect to the fiber axis, indicating the excitation of higher order LP_{11} mode alongside the fundamental LP_{01}.

The mode seen in Figure 2.13 right was associated with a transmission efficiency of 63%. By optimizing the alignment using the horizontal microscope the mode shape was optimized, and an efficiency of 80% has been achieved. For this optimization two screw of the mirror mounts were adjusted at the same time as a continuous walk-off, therefore performing an optimization remaining as long as possible in a good injection condition. The resulting mode is shown in Figure 2.14 left. Figure 2.14 right displays another mode obtained with nearly 80% of injection efficiency, where the surrounding kagome structure

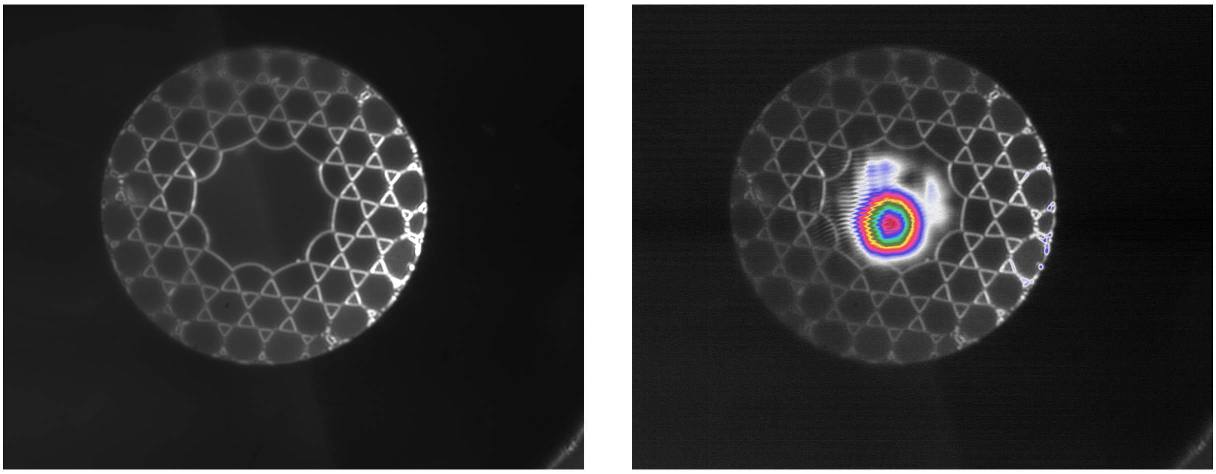

Figure 2.13: *Images taken with a preliminary horizontal microscope setup of the HCPCF kagome structure (left) and of the transmitted mode together with the kagome structure, with a pseudo color map (right). These images were taken with a preliminary setup of the horizontal microscope that provided a more homogeneous illumination, with the red light incident from two direction, making the kagome lattice distinctly visible.*

remains visible.

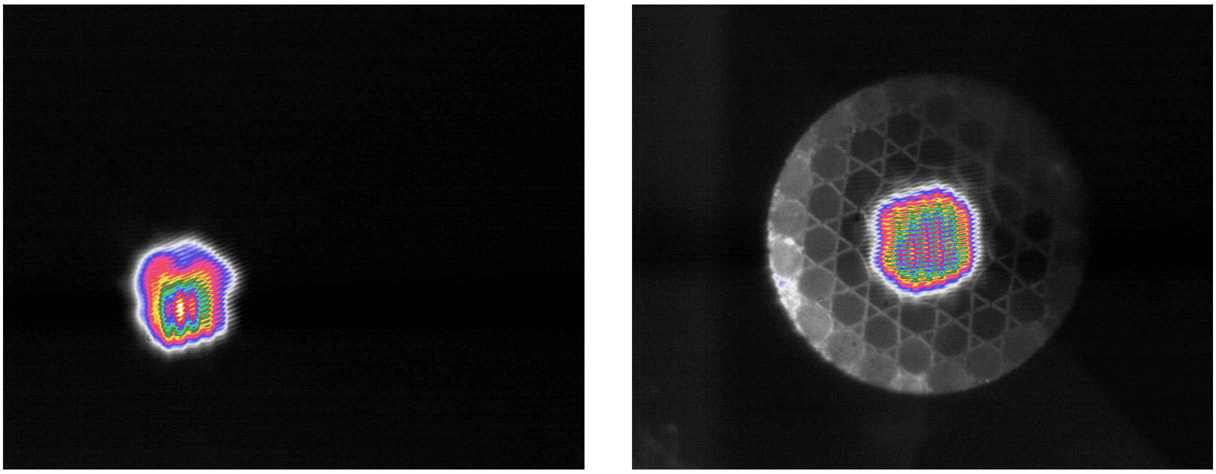

Figure 2.14: *Images captured with the final horizontal microscope setup. A mode with* $\sim$ *80% of transmission efficiency (left) and a guided mode with the kagome structure visible (right).*

A final optimization was performed without the ocular lens by translating the objective slightly past the working distance and placing the camera approximately 2 m away. With this configuration, the magnification achieved is about $\times 70$ (more than doubling the magnification of the horizontal microscope). At this distance, the light scattered from the kagome structure becomes very faint, complicating the focusing procedure as its signal approaches the noise level. However, employing the FLIR camera, which has a lower read noise and a high sensitivity, the kagome lattice remained resolvable, as shown in Figure 2.15. This figure shows clearly that the near-field mode assumes a hexagonal shape, perfectly fitting the core: as expected the field inherit the symmetry of the medium in which it is confined.

The advantage of this extended projection distance is the significant expansion of the mode, the pixel grid interference patterns disturb the shape far less, allowing finer spatial details to be resolved, further improving the alignment procedure. With this method an efficiency of 94% was achieved. The resulting mode (Figure 2.16 left) is still slightly asymmetric, indicating a residual superposition of higher order modes with the fundamental one.

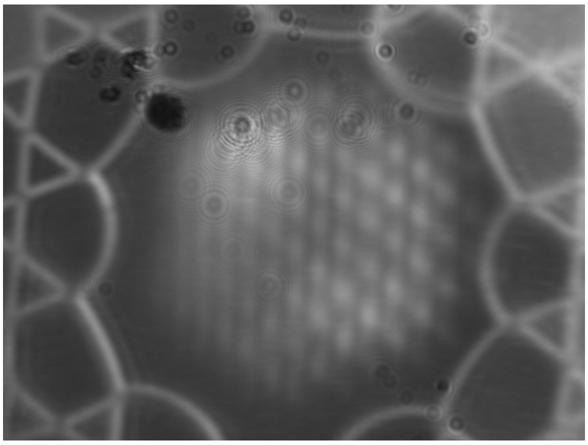

Figure 2.15: *Images obtained with the FLIR camera using the microscope objective without the ocular lens. The mode clearly exhibits a hexagonal shape. Thanks to the FLIR high sensitivity and low read noise, the kagome structure remains visible even in this configuration. Artifacts in the top-left corner are dust grain deposited on the camera sensor.*

Finally, a sweep of laser nominal power was conducted, which changes also the beam waist at the fiber injection. At every tested power the injection alignment was re-optimized, and the best mode profile was obtained at 6.5 W, with an efficiency of 96%. The mode is shown in Figure 2.16 right, in this case the shape is far more symmetric than the previous iterations, which, together with the high transmission efficiency, suggests a significant high share of the fundamental LP_{01} mode.

Qua manca qualcosa sulle due fotcamere, e anche un riferimento alla HCPCF (aggiungerò una sezione)

2.2.3.2 Chamber fiber

After validating the effectiveness of the alignment procedure using the test fiber, the same method was applied to the fiber installed inside the vacuum chamber. Unfortunately, it was not possible to properly illuminate the fiber, and the kagome structure could not be visualized. Using the same 635 nm LED employed for the test fiber, several configurations of focusing lenses and LED positions were tested, however, no light was successfully coupled into the kagome structures. In addition, due to physical constraints imposed by the vacuum chamber geometry, it was not possible to directly illuminate the fiber facet from the outside. One possible limitation might be the presence of a translucent plastic

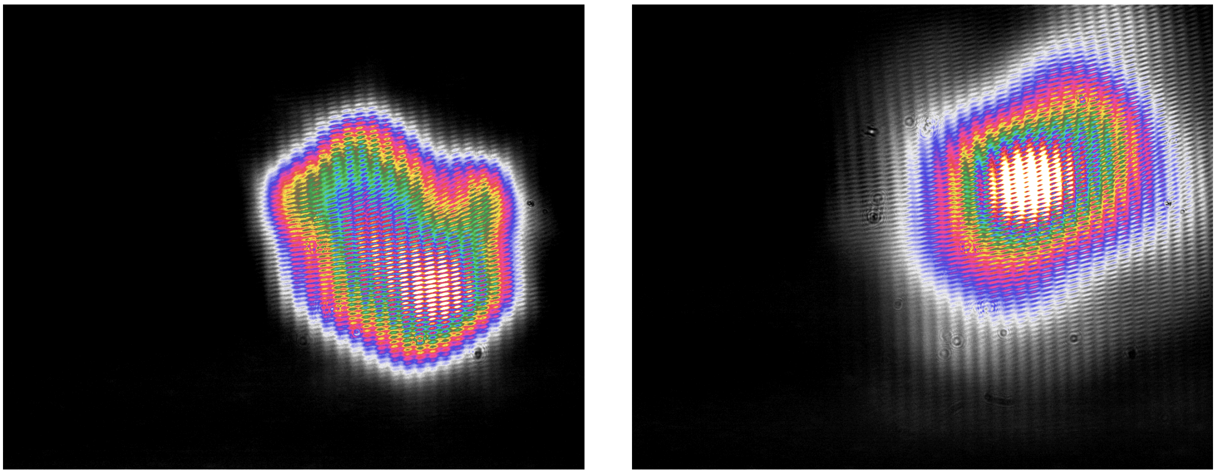

Figure 2.16: *Images obtained with the objective without the ocular lens. Mode with* 94% *of injection efficiency (left). Optimized mode with nominal laser power set to 6.5 W which achieved* 96% *of transmission efficiency (right).*

casing surrounding the fiber. The casing may diffuse the light of the LED preventing it to reach the kagome structure. Unlike the images obtained with the test fiber, in this case the jacket appears illuminated, suggesting that the light propagated in the jacket rather than reaching the fiber kagome structure.

Furthermore, the need to create a virtual image outside the chamber required the use of multiple lenses (see Section 2.2.2.3). While this approach enabled imaging of the fiber facet, it also introduced non-negligible chromatic aberrations. Figures 2.17 left and 2.17 right show respectively a preliminary image of the focused fiber and the same configuration with light guided through the fiber. In the latter case, the optical mode is in focus while the fiber facet appears out of focus.

The mode obtained after the injection alignment procedure is shown in Figure 2.18, which exhibited $\sim 50\%$ of transmission efficiency. It should be noted that this efficiency values was measured after the light propagated through a chamber window, two lenses and two mirrors, all of which introduce transmission and reflection losses. Moreover, the fiber installed in the chamber is longer than the test fiber. Although propagation losses associated with the additional length are should be negligible, the increased length requires several bends in the fiber routing, which are known to induce leakage losses in optical fibers.

The best mode obtained after the optimization procedure is displayed in Figure 2.18. It shows a clear asymmetry, indicating the presence of a superposition of various LP modes. Although, it should also be considered that, in order to enter the vacuum chamber, the fiber must be tightened in a ferrule by a swage-lock fitting. This mechanical stress can deform the fiber microstructure, altering its guiding properties and degrading the mode shape quality.

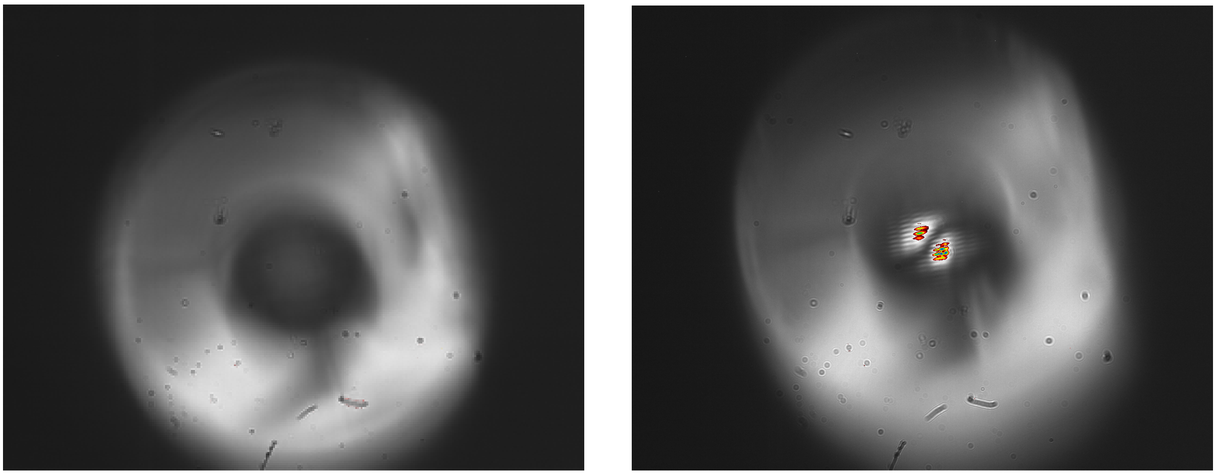

Figure 2.17: *Preliminary images taken with a zoom about* $\times 10$ *of the fiber facet in the vacuum chamber (left) and the fiber with a non-optimized mode (right). In contrast to the test fiber images, here, the kagome structure is not illuminated, whereas the outer jacket is. The mode exhibits a clear* LP_{11} *profile. Note that focusing the mode bring the fiber out of focus, this makes the mode seem non-centered with respect to the fiber core.*

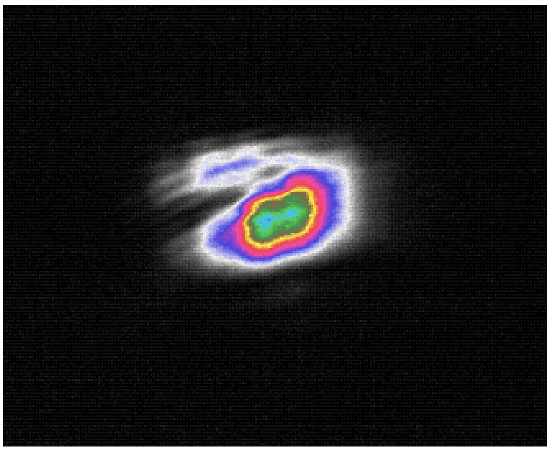

Figure 2.18: *Final mode profile obtained after the injection optimization (magnification about* $\times 10$*). The mode shape is a bit deformed, meaning that it is a superposition of fundamental LP modes. The image is cropped.*

2.2.4 ODT loading efficiency calculation

To have a theoretically calculated reference of the capture efficiency of the ODT, a small model has been developed in two regimes: shallow trap, in which the trap depth is assumed much smaller than the thermal energy, and the non-shallow trap in which the trap depth is assumed comparable to the thermal energy.

2.2.4.1 Model description

The ODT beam is modelled as a Gaussian beam propagating along the z-axis, its local intensity distribution as a function of the beam total power P is given by:

$$I(x,y,z) = \frac{2P}{\pi w(z)^2} \exp\left[-2\frac{x^2+y^2}{w(z)^2}\right], \tag{2.9}$$

where the beam radius $w(z)$ evolves along the propagation axis according to:

$$w(z) = w_0\sqrt{1+\left(\frac{z}{z_R}\right)^2}. \tag{2.10}$$

Using (2.9) in (1.55), the ODT potential energy can be expressed as

$$U_{\text{dip}}(\mathbf{r}) = \alpha\frac{2P}{\pi w(z)^2} \exp\left[-2\frac{x^2+y^2}{w(z)^2}\right], \tag{2.11}$$

where α represents the real part of the polarizability expressed in units of $1/(2\epsilon_0 c)$:

$$\alpha = -\frac{1}{2\epsilon_0 c}\text{Re}[\alpha'], \tag{2.12}$$

with α' being the polarizability defined as in Section 1.3.
The maximum trap depth U_0 is located at the beam waist w_0 position, which is assumed to be located at the fiber tip ($z=0$), and it is given by:

$$U_0 = \alpha I_{\max}, \tag{2.13}$$

where $I_{\max} = 2P/(\pi w_0^2)$ is the maximum intensity.
Following the molasses cooling stage, the atomic cloud is assumed in thermodynamic equilibrium, therefore, the phase space distribution is described by the spatial profile ($n_{\text{MOL}}(\mathbf{r})$) of the atomic cloud coupled with the Maxwell-Boltzmann velocity distribution:

$$f(\mathbf{r},\mathbf{v}) = n_{\text{MOL}}(\mathbf{r})\left(\frac{M}{2\pi k_B T}\right)^{3/2} \exp\left[-\frac{Mv^2}{2k_B T}\right], \tag{2.14}$$

where M is the mass of a ^{87}Rb atom, T is the sample temperature, and the total atom number is normalized via

$$N_{\mathrm{MOL}} = \int d^3r d^3v f(\mathbf{r}, \mathbf{v}). \tag{2.15}$$

An atom positioned at the spatial coordinate $\mathbf{r}$ is successfully trapped by the ODT if its transverse kinetic energy (in the x-y plane) is lower than the local depth of the trap:

$$v_x^2 + v_y^2 < \frac{2|U_{\mathrm{dip}}(\mathbf{r})|}{M}. \tag{2.16}$$

For the v_z velocity component, it is assumed that all the atoms having $v_z < 0$ are captured, eventually reaching the fiber tip.

Therefore, the total number of captured atoms N_{ODT} is evaluated by integrating over the phase-space volume of trapping success:

$$N_{\mathrm{ODT}} = \int d^3r\, n_{\mathrm{MOL}}(\mathbf{r}) \times \frac{1}{2} \iint_{v_x^2+v_y^2<\frac{2|U_{\mathrm{dip}}|}{M}} dv_x dv_y \left(\frac{M}{2\pi k_B T}\right) \exp\left[-\frac{M(v_x^2+v_y^2)}{2k_B T}\right], \tag{2.17}$$

where the factor 1/2 is introduced by the trapping criterion on the longitudinal component v_z of the velocity.

Passing to polar coordinates in the transverse velocity plane $v_r^2 = v_x^2 + v_y^2$, $dv_x dv_y = \pi d(v_r^2)$ and defining the dimensionless variable $x = (Mv_r^2)/(2k_BT)$, the velocity integral evaluates directly:

$$\frac{1}{2}\int_0^{\frac{|U_{\mathrm{dip}}(\mathbf{r})|}{k_BT}} dx\, e^{-x} = \frac{1}{2}\left(1 - \exp\left[-\frac{|U_{\mathrm{dip}}(\mathbf{r})|}{k_BT}\right]\right). \tag{2.18}$$

Consequently, the general expression for the ODT capture efficiency can be expressed as:

$$\frac{N_{\mathrm{ODT}}}{N_{\mathrm{MOL}}} = \frac{1}{2N_{\mathrm{MOL}}} \int d^3r\, n_{\mathrm{MOL}}(\mathbf{r}) \left(1 - \exp\left[-\frac{|U_{\mathrm{dip}}(\mathbf{r})|}{k_BT}\right]\right). \tag{2.19}$$

2.2.4.2 Shallow trap approximation

In the limit where the thermal energy of the atomic cloud is significantly greater than the local optical potential in the overlapping region ($|U_{\mathrm{dip}}(\mathbf{r})| \ll k_BT$), the exponential term can be linearized to first order:

$$1 - \exp\left(-\frac{|U_{\mathrm{dip}}(\mathbf{r})|}{k_BT}\right) \simeq \frac{|U_{\mathrm{dip}}(\mathbf{r})|}{k_BT}, \tag{2.20}$$

Equation (2.19) simplifies to:

$$N_{\mathrm{ODT}} \simeq \int d^3r\, n_{\mathrm{MOL}}(\mathbf{r}) \frac{|U_{\mathrm{dip}}(\mathbf{r})|}{2k_BT} = \frac{U_0}{2k_BT} \int d^3r\, n_{\mathrm{MOL}}(\mathbf{r}) \frac{w_0^2}{w(z)^2} \exp\left[-2\frac{x^2+y^2}{w(z)^2}\right]. \tag{2.21}$$

Since the transverse waist of the dipole beam ($w(z) \lesssim 100\ \mu$m) is significantly smaller than the transverse radii of the atomic cloud ($R_x, R_y \sim 1\,\mathrm{mm}$), the atomic density can be considered constant over the beam profile, and the integration along z can be decoupled from the x and y integration, obtaining:

$$\iint dxdy \, \exp\left[-2\frac{x^2+y^2}{w(z)^2}\right] = \frac{\pi w(z)^2}{2}. \tag{2.22}$$

Substituting this result in equation (2.21), the terms $w(z)^2$ cancel out, eliminating the explicit dependence on the beam divergence:

$$N_{\mathrm{ODT}} \simeq \frac{U_0}{2k_BT}\frac{\pi w_0^2}{2}\int dz \, n_{\mathrm{MOL}}(0,0,z). \tag{2.23}$$

Adopting a uniform-density ellipsoid model to represent the cloud, characterized by radii R_x, R_y, R_z and a constant core density

$$n_{\mathrm{MOL}} = \frac{3N_{\mathrm{MOL}}}{4\pi R_x R_y R_z}, \tag{2.24}$$

integration along the longitudinal axis over the ellipsoid boundaries ($\pm R_z$) yields the ODT capture efficiency:

$$\frac{N_{\mathrm{ODT}}}{N_{\mathrm{MOL}}} \simeq \frac{3}{8}\frac{U_0}{k_BT}\frac{w_0^2}{R_xR_y}. \tag{2.25}$$

In this regime, the loading efficiency is independent of the absolute distance between the cloud center and the fiber tip (d), and independent of the minimum waist w_0, since the product $U_0 w_0^2 = 2\alpha P/\pi$ depends solely on the total laser power. The primary experimental figure of merit to optimize is therefore the ratio $N_{\mathrm{MOL}}/(T \cdot R_xR_y)$.

2.2.4.3 Non-shallow trap

When the local trap depth becomes comparable to or greater than the kinetic energy of the atoms, the linear truncation introduces an overestimation. In this case the full Taylor series is needed:

$$1-\exp\left[-\frac{|U_{\mathrm{dip}}(\mathbf{r})|}{k_BT}\right] = \sum_{q=1}^{\infty}\frac{(-1)^{q+1}}{q!}\left[\frac{U_0}{k_BT}\frac{w_0^2}{w(z)^2}\exp\left(-2\frac{x^2+y^2}{w(z)^2}\right)\right]^q. \tag{2.26}$$

Under the same assumption of a highly localized beam relative to the cloud size ($w(z)/\sqrt{q} \ll R_x, R_y$), spatial integration over the transverse plane for the q-th order term yields:

$$\iint dxdy \, \exp\left[-2q\frac{x^2+y^2}{w(z)^2}\right] = \frac{\pi w(z)^2}{2q}. \tag{2.27}$$

Inserting this back into the volume integral yields the following series representation:

$$N_{\mathrm{ODT}} = \frac{1}{2}\sum_{q=1}^{\infty}\frac{(-1)^{q+1}\pi}{2q(q!)}\left(\frac{U_0}{k_B T}\right)^q w_0^2 \int dz\, n_{\mathrm{MOL}}(0,0,z)\left(\frac{w_0^2}{w(z)^2}\right)^{q-1}. \tag{2.28}$$

Using $w(z)^2/w_0^2 = 1 + z^2/z_R^2$, assuming constant density, and assuming the center of the atomic cloud is positioned at an axial displacement $z = d$ from the fiber tip, the longitudinal integral becomes:

$$\frac{3N_{\mathrm{MOL}}}{4\pi R_x R_y R_z}\int_{d-R_z}^{d+R_z} dz\left(1+\frac{z^2}{z_R^2}\right)^{1-q}. \tag{2.29}$$

Under the physical condition where the longitudinal radius of the cloud is significantly narrower than the Rayleigh range ($R_z \ll z_R$), the integrand remains approximately constant across the integration volume and can be evaluated at the cloud center $z = d$. This approximation leads to the result

$$\frac{3N_{\mathrm{MOL}}}{4\pi R_x R_y R_z}\int_{d-R_z}^{d+R_z} dz\left(1+\frac{z^2}{z_R^2}\right)^{1-q} = \frac{3N_{\mathrm{MOL}}}{2\pi R_x R_y}(1+d^2/z_R^2)^{1-q}. \tag{2.30}$$

Using (2.30) in (2.28), by defining the dimensionless parameter ξ, which evaluates the ratio of the trap depth at the cloud center to the thermal energy:

$$\xi = \frac{U_0}{k_B T\left(1+\frac{d^2}{z_R^2}\right)}, \tag{2.31}$$

the total captured atom fraction becomes:

$$\frac{N_{\mathrm{ODT}}}{N_{\mathrm{MOL}}} = \frac{3}{8}\frac{w_0^2}{R_x R_y}\left(1+\frac{d^2}{z_R^2}\right)\left[-\sum_{q=1}^{\infty}\frac{(-\xi)^q}{q\cdot q!}\right]. \tag{2.32}$$

The infinite series inside the brackets evaluates to (see Power Series section in [12])

$$-\sum_{q=1}^{\infty}\frac{(-\xi)^q}{q\cdot q!} = \gamma_E + \log(\xi) + \Gamma(0,\xi), \tag{2.33}$$

where γ_E is the Euler–Mascheroni constant. The final analytical solution for the non-shallow ODT capture efficiency is

$$\frac{N_{\mathrm{ODT}}}{N_{\mathrm{MOL}}} = \frac{3}{8}\frac{w_0^2}{R_x R_y}\left(1+\frac{d^2}{z_R^2}\right)\Big[\gamma_E + \log(\xi) + \Gamma(0,\xi)\Big]. \tag{2.34}$$

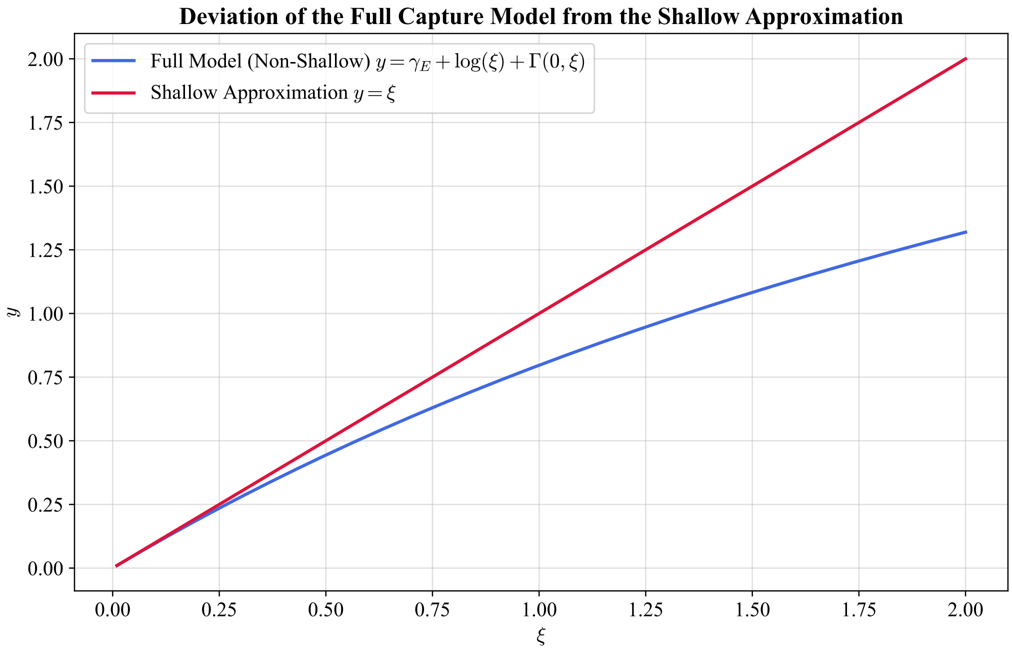

Figure 2.19: *Comparison between the relative ODT capture efficiency calculated using the shallow* $y = \xi$ *(red line) and non-shallow* $y = \gamma_E + \log(\xi) + \Gamma(0, \xi)$ *(blue line) models.*

The shallow trap result (2.25) is recovered by the substitution:

$$\gamma_E + \log(\xi) + \Gamma(0, \xi) \to \xi, \tag{2.35}$$

corresponding also to the limit $\xi \to 0$. The comparison between the models is shown in Figure 2.19.

2.3 Experimental Procedures and analysis methodology

The first step of the experimental sequence is the capture of atoms in the MOT, using the optimal parameters previously determined in for this apparatus [13]. The MOT is realized with an axial magnetic field gradient of $B' = 9.32$ G/cm (corresponding to a current of 8 A), a cooler laser detuning of $\delta = -13$ MHz, and all compensation coil currents set to 0 mA. The MOT is loaded for a total duration of 5 s.
After the loading phase, the magnetic field is switched off, initiating the optical molasses phase. In the absence of the magnetic field, the atomic temperature decreases, however the confinement force is lost, and the atomic cloud begins to expand. At the same time, under the influence of gravity, the cloud starts to fall while experiencing a viscous-like damping force due to its interaction with the laser light. The temperature achieved during the molasses phase scales as $I/|\delta|$, where I is the laser intensity and δ is the detuning [2]. However, this relation remains valid as long as $|\delta|$ is sufficiently small to not reduce significantly the optical cycle rate $\Gamma_{\mathrm{p}} \propto 1/\delta^2$. Additionally, the detuning must not be so large to induce a significant excitation probability of the neighboring $F = 2 \to F' = 2$ transition. Furthermore, optimal molasses performance is obtained when the magnetic field is effectively zero. For this reason 5 ms after the MOT magnetic field is switched off, the shim coil currents are adjusted to compensate for the Earth's magnetic field and any residual magnetic fields present in the experimental environment. The molasses is performed with a detuning different from that used for the MOT phase, it is changed with a ramp of 20 steps with a total duration of 5 ms. The ramp stars 50 μs after the shim coils are changed to their molasses value.

2.3.1 Atom Imaging

Two different CMOS cameras were used to image the atoms inside the vacuum chamber:

- **Teledyne FLIR BFS-U3-04S2M-CS**, monochromatic camera mounting the Sony IMX287 sensor, with a resolution of 720×540, and a pixel size of 6.9 μm $\times$ 6.9 μm.

- **Allied Vision Manta G-419 NIR**, monochromatic camera mounting the CMOSIS/ams CMV4000 CMOS sensor, with a resolution of 2048×2048, and a pixel size of 5.5 μm $\times$ 5.5 μm.

The FLIR camera was primarily used during the optimization of the molasses phase, whereas, the Manta camera was employed for measurements of the atom capture efficiency of the ODT. The Manta was preferred with respect to the FLIR because of its higher

sensitivity at 780 nm and its higher pixel density, which helped the imaging of the small region where the ODT is present.
Both cameras were equipped with a Thorlabs MVL50M23 adjustable objective lens with a focal length of 50 mm, and a minimum working distance of 200 mm, corresponding to a minimum magnification of ~ 0.33. The objective was adjusted to focus on the HCPCF inside the vacuum chamber, ensuring that the atomic cloud, located at a similar distance, was also in focus. Both cameras were externally triggered by signals sent by the FPGA control system.
Since the temperature determination of the atomic cloud requires knowledge of its physical dimensions (see Section 2.3.2), the FLIR camera had to be calibrated.
The first calibration method was based on the measure of the diameter in pixel of the fiber inside the vacuum chamber. By turning on the MOT beam, the fiber gets illuminated, and a picture is taken. Using this method the fiber diameter of ~ 316 μm was found to correspond to 8 pixels. This yielded a calibration factor of 39.5 μm/px, corresponding to a magnification of approximately 0.17. However, the measurement uncertainty was significant: assuming an uncertainty of one pixel on each edge of the fiber image, the measured diameter becomes (8 ± 2) px, resulting in a calibration factor of (40 ± 10) μm/px. Consequently, this method provides only a rough estimate of the imaging magnification.
A second and more accurate calibration method was therefore employed. A mirror was positioned in front of the camera without modifying the objective setting, and a ruler with a visible scale was positioned so that it was in focus. This ensured that the ruler was located at the same distance from the objective as the fiber inside the vacuum chamber. With this method 2 cm corresponded to (471 ± 2) px. Since the instrumental uncertainty associated with a single measurement was negligible, the calibration was performed by taking five independent measurement, with the ruler repositioned at focus each time. This procedure yielded a final calibration factor of (43.4 ± 0.3) μm/px, corresponding to a magnification of 0.16.

2.3.2 Temperature measurements

Temperature measurements were performed to optimize the molasses parameters, with the goal of achieving the lowest possible temperature.
The temperature of the atomic sample was determined by measuring its ballistic expansion during free fall. Assuming a non-interacting gas and considering a single spatial coordinate, if an atom is found at x_0 at $t = 0$, after a free expansion time t its position is given by

$$x(t) = x_0 + v_x t. \tag{2.36}$$

Taking into account the distributions of both positions and velocities, and assuming that these quantities are statistically independent, the variance of their sum is equal to the sum of their variances. Therefore,

$$\sigma_x^2(t) = \sigma_{x,0}^2 + \sigma_{v_x}^2 t^2, \tag{2.37}$$

where $\sigma_{x,0}$ is the standard deviation of the position distribution at $t = 0$.
According to classical energy equipartition theorem, the temperature associated with a given spatial direction is related to the squared mean velocity along that direction:

$$\frac{1}{2} m \langle v_x^2 \rangle = \frac{1}{2} k_B T_x, \tag{2.38}$$

where m is the atomic mass and k_B is the Boltzmann constant.
In the frame of reference of the cloud center of mass, the atomic cloud is globally at rest, thus $\langle v_x \rangle = 0$ and

$$\langle v_x^2 \rangle = \sigma_{v_x}^2 = \frac{k_B T_x}{m}. \tag{2.39}$$

Substituting this result into 2.37 one gets:

$$\sigma_x^2(t) = \sigma_{x,0}^2 + \frac{k_B T_x}{m} t^2. \tag{2.40}$$

Using the atomic mass of the ^{87}Rb

$$m = m_A(^{87}R) = 86.9 \text{ u} \approx 1.443 \times 10^{-25} \text{ kg}, \tag{2.41}$$

and acquiring images at different times of flight (TOF), equation (2.40) can be fitted obtaining T_x as a fitting parameter.
The complete acquisition and analysis procedure is summarized below:

1. First a MOT is loaded, followed by the molasses phase. For the molasses different parameters and durations were tested in order to find the best configuration that minimized the temperature (see Section 3.1)

2. After the molasses stage, all cooling beams are switched off, allowing the atomic cloud to expand and fall freely. Using the FLIR camera images are taken at different TOFs, obtaining the intensity distribution $I(TOF)$. For each TOF a corresponding background image $I_{\text{bkg}}(TOF)$ is also acquired. The background image is taken by waiting a sufficiently long time (1s) ensuring that the cloud has completely disappeared and only the background signal remains.

   It should be noted that, due to the time required by the FLIR camera to transfer an image to the computer, it is not possible to take multiple TOF image during the same experimental run, therefore each image requires a new MOT loading and molasses sequence.

3. The background is subtracted

$$I_{\text{clean}}(TOF) = I(TOF) - I_{\text{bkg}}(TOF), \tag{2.42}$$

and the associated Poisson uncertainty is calculated as

$$\sigma_{\text{N}} = \sqrt{I(TOF) + I_{\text{bkg}}(TOF)}. \tag{2.43}$$

4. A two-dimensional Gaussian is fitted to each image (TRF minimization algorithm) to extract the cloud widths σ_x and σ_y along the two principal axis.

5. Equation (2.40) is fitted independently for both directions, obtaining the temperatures T_x and T_y. Then, the final temperature T is calculated as their average.

2.3.3 ODT loading measurements

In the analysis of the ODT, two main figures of merits are taking into account, namely the loading efficiency η and the intensity integral of the ODT peak, that is directly proportional to the number of captured atoms. They are measured by comparing fluorescence images acquired with the ODT either present or absent during the MOT and molasses stages. Since only a small fraction (order of 1/1000) of the atoms is expected to be captured by the ODT, the analysis relies on differential imaging and careful background normalization to isolate the signal of the trapped atoms from fluctuations in the total atom number. The complete analysis procedure is summarized below.

1. Two identical experimental sequences of MOT and molasses are consecutively performed with the same parameters. In one sequence, the ODT is kept on throughout the entire cycle starting from the MOT loading phase, whereas in the other the ODT remains off. After a TOF of 30 ms, fluorescence images of the atomic distribution denoted by I_{on} and I_{off} are taken with the Manta camera. During the TOF the atom not trapped in the ODT expands enhancing the visibility of those confined in the ODT. Each pair of measurements is repeated five times

2. These images are cropped to a region of interest of 530×380 pixels centered on the atomic cloud. This removes the region where the HCPCF is visible as well as peripheral areas containing only background signal.

3. To improve the signal-to-noise ratio, the images corresponding to each experimental condition are averaged:

$$\bar{I}\text{on} = \langle I_{\text{on}} \rangle \qquad \bar{I}\text{off} = \langle I_{\text{off}} \rangle \, .$$

This is important because the number of atom loaded into the ODT is expected to be small.

4. The averaged images are converted into one dimensional density profiles by integrating along the columns. This further increase the signal-to-noise ration.

5. The total number of atoms in the molasses exhibits slow fluctuations even when the experimental parameters are unchanged. These variations are attributed to slow drifts in the chamber pressure and to thermal effects in optical components, such as the AOMs. Since the ODT signal is small, due to the correlated nature of the drifts, despite the average of different images the difference would result in a background different from zero that is comparable to the ODT signal amplitude. To compensate this effect, a rescaling procedure is applied to the background images. This method computes a scale factor, k, which enforces a zero integrated difference outside the ODT region.

   The procedure is implemented as follows:

   i. An exclusion region centered on the ODT position, $x_{\mathrm{ODT}} \pm R_{\mathrm{window}}$ is defined. All remaining positions constitute the outside region.

   ii. A linear background $L_i(x)$ is estimated by interpolating the edges of each intensity profile. After subtracting the background, the integrated signal outside the ODT region is computed for both the ODT-on and ODT-off profiles ($A^{\mathrm{out}}_{\mathrm{on}}$ and $A^{\mathrm{out}}_{\mathrm{off}}$) as

$$A_i^{\mathrm{out}} = \int_{\mathrm{outside}} \Big(P_i(x) - \bar{I}_i(x)\Big)\, dx \qquad i \in \{\mathrm{on}, \mathrm{off}\} \tag{2.44}$$

   iii. The normalization factor is given by the ratio of these integrated areas:

$$k = \frac{A^{\mathrm{out}}_{\mathrm{on}}}{A^{\mathrm{out}}_{\mathrm{off}}} \tag{2.45}$$

6. The final rescaled profile, which isolates the signal of the atoms trapped in the ODT, is obtained as:

$$\Delta\bar{I}(x) = \bar{I}_{\mathrm{on}}(x) - k \cdot \bar{I}_{\mathrm{off}}(x) \tag{2.46}$$

7. Two independent fits consisting of a Gaussian function and a linear background are performed. The first fit is applied to the ODT-off profile $\bar{I}_{\mathrm{off}}(x)$,while the second is applied to the differential profile $\Delta\bar{I}(x)$.
   The areas $A^{\mathrm{Gaus}}_{\mathrm{off}}$ and $A^{\mathrm{Gaus}}_{\mathrm{ODT}}$ are calculated from the fit results, and the capture efficiency is calculated as

$$\eta = \frac{A^{\mathrm{Gaus}}_{\mathrm{ODT}}}{k \times A^{\mathrm{Gaus}}_{\mathrm{off}}} \tag{2.47}$$

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